

A1 Number — Lesson Plan

IGCSE Mathematics 0580 | Concept Guide

A1_Q1: Number Types

Hook

Ever wondered why your phone's calculator gives you an error when you try to divide by zero? Or why some numbers like π never seem to end? The way we classify numbers determines what operations are possible — and some numbers have been so controversial that mathematicians were executed for discovering them!

Prerequisites

- Basic arithmetic operations (addition, subtraction, multiplication, division)
- Understanding of place value and decimal notation
- Familiarity with fractions as parts of a whole

Core Explanation

Numbers are classified into a hierarchy of sets, each building on the one before.

Natural Numbers $\mathbb{N} = \{1, 2, 3, 4, \dots\}$ are the counting numbers. Some definitions include 0, but in IGCSE we usually start at 1. These are the first numbers humans ever used.

Integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ extend naturals to include negatives and zero. The letter $\mathbb{Z}$ comes from the German word **Zahlen** meaning 'numbers'. Integers are closed under addition, subtraction, and multiplication — meaning any integer operation on integers yields another integer.

Rational Numbers $\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$ are numbers that can be written as a fraction of two integers. This includes whole numbers (like $\frac{4}{1}$), terminating decimals (like $0.75 = \frac{3}{4}$), and recurring decimals (like $0.\dot{3} = \frac{1}{3}$). The letter $\mathbb{Q}$ stands for 'quotient'.

Irrational Numbers are numbers that **cannot** be written as a fraction of two integers. Their decimal expansions go on forever without repeating. Famous examples include:

- $\pi = 3.14159265\dots$
- $\sqrt{2} = 1.41421356\dots$
- $e = 2.71828182\dots$

Real Numbers $\mathbb{R}$ include **all** rational and irrational numbers together. They fill the number line completely with no gaps.

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

Every natural number is an integer. Every integer is rational. Every rational is real. But not every real number is rational!

Worked Example

Problem: Classify each number as natural, integer, rational, irrational, or real. List all sets that apply.

Numbers: -3 , $\frac{2}{5}$, $\sqrt{16}$, π , 0

Solution:

1. -3

- Integer? Yes ($-3 \in \mathbb{Z}$)
- Rational? Yes ($-3 = \frac{-3}{1}$)
- Real? Yes
- Natural? No (natural numbers are positive)
- **Answer:** Integer, rational, real

2. $\frac{2}{5}$

- Rational? Yes (it's a fraction of two integers)
- Real? Yes
- Integer? No
- **Answer:** Rational, real

3. $\sqrt{16}$

- Simplify: $\sqrt{16} = 4$
- Natural? Yes ($4 \in \mathbb{N}$)
- Integer, rational, real? Yes to all
- **Answer:** Natural, integer, rational, real

4. π

- Rational? No (cannot be written as $\frac{a}{b}$)
- Irrational? Yes
- Real? Yes
- **Answer:** Irrational, real

5. 0

- Integer? Yes
- Rational? Yes ($0 = \frac{0}{1}$)
- Real? Yes
- Natural? No (usually; check your syllabus definition)
- **Answer:** Integer, rational, real

Common Mistakes

X Assuming all square roots are irrational (e.g., saying $\sqrt{9}$ is irrational)

OK Fix: Always simplify first! $\sqrt{9} = 3$, which is rational. Only square roots of non-perfect squares (like $\sqrt{2}$) are irrational.

X Confusing rational numbers with fractions — thinking $\frac{\pi}{2}$ is rational because it looks like a fraction

OK Fix: A rational number must be $\frac{a}{b}$ where **both** a and b are integers. Since π is not an integer, $\frac{\pi}{2}$ is not rational.

X Thinking that 0 is not an integer

OK Fix: Zero is definitely an integer. It's the integer that separates positive from negative numbers. Check your number line!

Summary

- Natural numbers $\mathbb{N}$ are the counting numbers (1, 2, 3, ...)
- Integers $\mathbb{Z}$ include all whole numbers — positives, negatives, and zero
- Rational numbers $\mathbb{Q}$ can be written as $\frac{a}{b}$ where a, b are integers and $b \neq 0$
- Irrational numbers cannot be written as fractions — their decimals never terminate or repeat
- Real numbers $\mathbb{R}$ include all rational and irrational numbers
- The hierarchy is: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$
- Always simplify expressions before classifying

A1_Q2: Factors, Multiples & Primes

Hook

The RSA encryption that protects your WhatsApp messages, online banking, and even top-secret military communications relies on one simple fact: multiplying two large primes is easy, but finding the prime factors of the product is incredibly hard — even for supercomputers. Without prime numbers, modern cryptography wouldn't exist!

Prerequisites

- Multiplication and division facts up to 12×12
- Understanding of integer arithmetic
- Basic concept of even and odd numbers

Core Explanation

Factors (or divisors) of a number are whole numbers that divide it exactly with no remainder. For example, the factors of 12 are 1, 2, 3, 4, 6, 12. Every number has at least two factors: 1 and itself.

Multiples of a number are what you get when you multiply it by any integer. The multiples of 5 are 5, 10, 15, 20, ... (i.e., $5 \times 1, 5 \times 2, 5 \times 3, \dots$).

Prime numbers have exactly two distinct factors: 1 and themselves. The first few primes are 2, 3, 5, 7, 11, 13, 17, 19, 23, ... Note that 2 is the **only** even prime — every other even number has 2 as a factor.

Composite numbers have more than two factors (e.g., 4, 6, 8, 9, 10, \dots).

Prime Factorisation is expressing a number as a product of its prime factors. The standard method uses a factor tree:

For 60:

$$60 = 6 \times 10 = (2 \times 3) \times (2 \times 5) = 2^2 \times 3 \times 5$$

Highest Common Factor (HCF) is the largest factor shared by two or more numbers.

Lowest Common Multiple (LCM) is the smallest positive multiple shared by two or more numbers.

Method using prime factors: To find HCF and LCM of 60 and 84:

$$60 = 2^2 \times 3 \times 5$$

$$84 = 2^2 \times 3 \times 7$$

- **HCF:** take each prime to the **smallest** power that appears: $2^2 \times 3 = 12$
- **LCM:** take each prime to the **largest** power that appears: $2^2 \times 3 \times 5 \times 7 = 420$

Divisibility Rules help you check without long division:

- Divisible by 2: last digit is even
- Divisible by 3: sum of digits is divisible by 3
- Divisible by 4: last two digits form a number divisible by 4
- Divisible by 5: last digit is 0 or 5
- Divisible by 6: divisible by both 2 and 3
- Divisible by 9: sum of digits is divisible by 9
- Divisible by 10: last digit is 0

Worked Example

Problem: A gardener has two lengths of rope: 84 cm and 108 cm. She wants to cut them into equal-length pieces, as long as possible, with no rope wasted. Find the length of each piece and the total number of pieces.

Solution:

The longest possible piece length is the HCF of 84 and 108.

Step 1: Prime factorise each number.

$$84 = 2 \times 42 = 2 \times 6 \times 7 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$$

$$108 = 2 \times 54 = 2 \times 6 \times 9 = 2 \times 2 \times 3 \times 3 \times 3 = 2^2 \times 3^3$$

Step 2: HCF = common primes with smallest powers.

Common primes: 2 and 3. Smallest powers: 2^2 and 3^1 .

$$\text{HCF} = 2^2 \times 3 = 4 \times 3 = 12$$

Each piece is 12 cm long.

Step 3: Number of pieces.

First rope: $84 \div 12 = 7$ pieces

Second rope: $108 \div 12 = 9$ pieces

Total pieces: $7 + 9 = 16$ pieces

Answer: She cuts 12 cm pieces and gets 16 pieces in total.

Common Mistakes

X Confusing HCF and LCM — using largest powers when finding HCF or smallest powers when finding LCM

OK Fix: Remember: **HCF** uses the **H**ighest factor that's **C**ommon — take **s**mallest powers. **LCM** uses the **L**argest factor that's **C**ommon — take **l**argest powers.

X Forgetting that 1 is neither prime nor composite

OK Fix: 1 has only one factor (itself), so it doesn't satisfy the definition of prime (exactly two factors) or composite (more than two factors).

X Thinking 1 is a prime number because it can only be divided by 1 and itself (which is also 1)

OK Fix: A prime must have **exactly two distinct** positive factors. 1 has only one positive factor, so it is not prime.

Summary

- Factors divide a number exactly; multiples are the number times any integer
- Primes have exactly two factors: 1 and themselves
- Prime factorisation expresses a number as a product of prime powers
- HCF uses smallest powers of common primes; LCM uses largest powers of all primes
- Divisibility rules save time when checking factors
- Real-world applications include rope cutting, scheduling, and encryption

A1_Q3: Fractions & Decimals

Hook

In ancient Egypt, they had a completely different way of writing fractions — they only used unit fractions like $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{7}$, and would express $\frac{3}{4}$ as $\frac{1}{2} + \frac{1}{4}$. Imagine doing your maths homework with that system! Our decimal system, while familiar, hides the fact that some innocent-looking fractions like $\frac{1}{3}$ go on forever as $0.3333\dots$

Prerequisites

- Basic arithmetic operations (+, −, ×, ÷)

- Understanding of place value in decimals
- Multiplication tables up to 12×12

Core Explanation

Fractions represent parts of a whole. The **numerator** (top) shows how many parts we have; the **denominator** (bottom) shows how many equal parts the whole is divided into.

Operations with Fractions

Addition and Subtraction: You need a **common denominator**.

$$\frac{2}{5} + \frac{1}{3} = \frac{2 \times 3}{5 \times 3} + \frac{1 \times 5}{3 \times 5} = \frac{6}{15} + \frac{5}{15} = \frac{11}{15}$$

Multiplication: Multiply numerators together and denominators together.

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$

Division: Multiply by the reciprocal (flip the second fraction).

$$\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{10}{12} = \frac{5}{6}$$

Recurring Decimals

Some fractions produce decimals that repeat forever. The repeating part is shown with a dot (or dots) above:

- $\frac{1}{3} = 0.\dot{3}$ (repeats every 1 digit)
- $\frac{1}{6} = 0.1\dot{6}$ (the 6 repeats)
- $\frac{1}{7} = 0.\dot{1}4285\dot{7}$ (repeats every 6 digits)

Converting Recurring Decimals to Fractions

Let $x = 0.\dot{7}$ (meaning $0.7777\dots$).

Multiply by 10: $10x = 7.\dot{7}$

Subtract: $10x - x = 7.\dot{7} - 0.\dot{7}$

$$9x = 7$$

$$x = \frac{7}{9}$$

For $0.1\dot{6}$ (meaning $0.1666\dots$):

$$x = 0.1666\dots$$

$$10x = 1.666\dots$$

$$100x = 16.666\dots$$

$$100x - 10x = 16.666\dots - 1.666\dots = 15$$

$$90x = 15$$

$$x = \frac{15}{90} = \frac{1}{6}$$

Mixed Numbers and Improper Fractions: A mixed number like $2\frac{3}{5}$ means $2 + \frac{3}{5} = \frac{10}{5} + \frac{3}{5} = \frac{13}{5}$.

Worked Example

Problem: Calculate $\frac{3}{4} - \frac{2}{3} \times \frac{1}{2}$ and express your answer as a fraction in its simplest form.

Solution:

Remember BIDMAS (or PEMDAS) — multiplication before subtraction.

Step 1: Do the multiplication first.

$$\frac{2}{3} \times \frac{1}{2} = \frac{2 \times 1}{3 \times 2} = \frac{2}{6} = \frac{1}{3}$$

Step 2: Now subtract.

$$\frac{3}{4} - \frac{1}{3}$$

Step 3: Find a common denominator. LCM of 4 and 3 is 12.

$$\frac{3 \times 3}{4 \times 3} - \frac{1 \times 4}{3 \times 4} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}$$

Answer: $\frac{5}{12}$

Check: $\frac{3}{4} = 0.75$, $\frac{1}{3} \approx 0.333$, $0.75 - 0.333 = 0.417$, $\frac{5}{12} \approx 0.4167$. ✓

Common Mistakes

X Adding denominators when adding fractions: $\frac{1}{3} + \frac{1}{4} = \frac{2}{7}$

OK Fix: Never add denominators! You must find a common denominator first: $\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$

X Misapplying 'keep-change-flip' for division and getting the first fraction flipped instead

OK Fix: Only flip (take the reciprocal of) the **second** fraction. $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{10}{12} = \frac{5}{6}$

X Forgetting to multiply the numerator when converting a mixed number to an improper fraction:

OK Fix: Multiply the whole number by the denominator, then add the numerator: $a\frac{b}{c} = \frac{a \times c + b}{c}$

Summary

- Add/subtract fractions by finding a common denominator
- Multiply fractions straight across: $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$
- Divide fractions by multiplying by the reciprocal of the second fraction
- Always simplify fractions to lowest terms
- Convert recurring decimals to fractions using the subtraction method
- Follow BIDMAS/PEMDAS for mixed operations

A1_Q4: Percentages

Hook

If you put \$1000 in a savings account earning 5% compound interest per year, after 40 years you'd have over \$7000 — without lifting a finger! But during a sale, that '50% off' tag is often followed by an extra '10% off' — is that 60% off total? (Spoiler: it's not.) Understanding percentages is the single most useful maths skill for everyday life.

Prerequisites

- Fractions and decimal operations
- Basic multiplication and division
- Converting between fractions, decimals, and percentages

Core Explanation

A **percentage** means 'out of 100'. $45\% = \frac{45}{100} = 0.45$.

Percentage of a Quantity

To find $p\%$ of a quantity, multiply by $\frac{p}{100}$:

$$15\% \text{ of } 80 = \frac{15}{100} \times 80 = 0.15 \times 80 = 12$$

Percentage Change

Percentage increase or decrease is calculated as:

$$\text{Percentage change} = \frac{\text{New value} - \text{Original value}}{\text{Original value}} \times 100\%$$

If a phone costing 200 is reduced to 170:

$$\frac{170 - 200}{200} \times 100\% = \frac{-30}{200} \times 100\% = -15\%$$

So it's a 15% decrease.

Reverse Percentages

Sometimes you know the final value after a percentage change and need to find the original. This is where **multipliers** help:

A 15% increase means multiply by $1 + \frac{15}{100} = 1.15$

A 15% decrease means multiply by $1 - \frac{15}{100} = 0.85$

If a laptop costs 690 after a 15% discount, the original price x satisfies:

$$0.85x = 690$$

$$x = \frac{690}{0.85} = 811.76$$

So the original price was 811.76.

Compound Interest

Compound interest means earning interest on previously earned interest. The formula is:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Where:

- A = final amount
- P = principal (initial amount)
- r = annual interest rate (as a decimal)
- n = number of times interest is compounded per year
- t = time in years

For IGCSE, a simplified version is often used:

$$A = P\left(1 + \frac{r}{100}\right)^t$$

where r is the annual percentage rate and compounding is yearly.

Example: £2000 invested at 4% per year for 3 years:

$$A = 2000\left(1 + \frac{4}{100}\right)^3 = 2000 \times (1.04)^3 = 2000 \times 1.124864 = £2249.73$$

Worked Example

Problem: A house was bought for £250000. After 2 years, it was worth £280900. The owner claims the house appreciated at 6% per year. Is this correct?

Solution:

Step 1: Calculate what the value would be after 2 years at 6% per year.

$$A = 250000 \times (1.06)^2$$

Step 2: Simplify.

$$(1.06)^2 = 1.1236$$

$$250000 \times 1.1236 = 280900$$

Step 3: Compare with given value.

The calculated value £280900 matches the given value.

Answer: Yes, the owner is correct. The house appreciated at exactly 6% per year.

Alternative check: Find the actual annual growth rate.

$$\frac{280900}{250000} = 1.1236$$

$$\sqrt{1.1236} = 1.06$$

$$1.06 - 1 = 0.06 = 6\%$$

Common Mistakes

X Adding percentage changes: thinking two 10% increases equal a 20% increase

OK Fix: A 10% increase followed by another 10% increase is $1.1 \times 1.1 = 1.21$, which is a 21% increase.

X In reverse percentage problems, subtracting the percentage from the final value instead of dividing

OK Fix: If you know the price after a 20% increase is \$60, don't do $60 - 12 = 48$. Instead: original $= \frac{60}{1.2} = 50$.

X Using simple interest formula when compound interest is called for, or vice versa

OK Fix: Simple interest: same amount each year ($I = Prt$). Compound interest: interest on interest ($A = P(1 + \frac{r}{100})^t$)

Summary

- A percentage is a fraction out of 100
- Percentage change = $\frac{\text{change}}{\text{original}} \times 100\%$
- Use multipliers: $1 + \frac{r}{100}$ for increase, $1 - \frac{r}{100}$ for decrease
- Reverse percentages: divide by the multiplier to find the original
- Compound interest: $A = P(1 + \frac{r}{100})^t$
- Percentage changes multiply, not add

A1_Q5: Rounding & Estimation

Hook

A bridge collapses because an engineer rounded 6.58 metres to 7 metres — that 42 cm difference meant the steel beams didn't fit. In 1999, NASA lost a \$125 million Mars orbiter because one team used metric units and another used imperial. Rounding and precision aren't just classroom exercises — lives and billions of dollars depend on getting them right.

Prerequisites

- Place value understanding (units, tens, hundreds, tenths, hundredths)
- Basic arithmetic operations
- Understanding of decimal numbers

Core Explanation

Decimal Places (d.p.)

Rounding to a given number of decimal places:

1. Count the required number of digits after the decimal point
2. Look at the **next** digit (the one after your target)
3. If it's 5 or more, round **up**; if it's 4 or less, round **down**

Round 3.14159 to 2 decimal places:

- Look at the third decimal digit: 9
- $9 \geq 5$, so round the second decimal up
- $3.14159 \approx 3.14$ (actually 3.142 since $9 \geq 5$ means we round up the 1 to 2)

Wait — let me redo that carefully:

3.14159

- 1st d.p.: 3.1
- 2nd d.p.: look at digit in 3rd place = 1. $1 < 5$, so 3.14

- 3rd d.p.: look at digit in 4th place = 5. 5 $\geq$ 5, so 3.142

Significant Figures (s.f.)

The first significant figure is the first **non-zero** digit from the left.

- 0.004 has 1 significant figure (the 4)
- 0.0040 has 2 significant figures (the 4 and the 0)
- 5200 has 2 significant figures (the 5 and 2; trailing zeros without decimal point are ambiguous)
- 5200. has 4 significant figures (the decimal point makes trailing zeros significant)

Round 20458 to 2 significant figures:

- First s.f. is 2, second s.f. is 0
- Look at the third digit: 4
- $4 < 5$, so don't round up
- $20458 \approx 20000$ (to 2 s.f.)

Bounds and Error Intervals

When a measurement is rounded, the **true value** lies within an interval. If a length is 7.3 cm to 1 decimal place, then:

$$7.25 \leq \text{true value} < 7.35$$

The **lower bound** is 7.25 and the **upper bound** is 7.35.

For calculations with bounds:

- **Addition:** Lower bound = $LB_1 + LB_2$, Upper bound = $UB_1 + UB_2$
- **Subtraction:** To maximise $A - B$: use UB of A - LB of B. To minimise: use LB of A - UB of B.
- **Multiplication:** Lower bound = $LB_1 \times LB_2$, Upper bound = $UB_1 \times UB_2$
- **Division:** To maximise $A \div B$: use UB of A $\div$ LB of B. To minimise: use LB of A $\div$ UB of B.

Estimation

Estimate by rounding each number to 1 significant figure:

$$\frac{49.3 \times 6.12}{0.197} \approx \frac{50 \times 6}{0.2} = \frac{300}{0.2} = 1500$$

The actual answer is 1531.6... — the estimate is close enough to check your calculation is reasonable.

Worked Example

Problem: A rectangle is measured as 12.4 cm by 8.7 cm, each to 1 decimal place.

- Write the lower and upper bounds for each measurement.
- Find the lower and upper bounds for the area of the rectangle.

Solution:

(a) For 12.4 cm (to 1 d.p.):

Lower bound: 12.35 cm

Upper bound: 12.45 cm

For 8.7 cm (to 1 d.p.):

Lower bound: 8.65 cm

Upper bound: 8.75 cm

$$\begin{aligned} \text{(b) Lower bound of area} &= \text{LB}(\text{length}) \times \text{LB}(\text{width}) \\ &= 12.35 \times 8.65 = 106.8275 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Upper bound of area} &= \text{UB}(\text{length}) \times \text{UB}(\text{width}) \\ &= 12.45 \times 8.75 = 108.9375 \text{ cm}^2 \end{aligned}$$

****Answer:**** The area lies between 106.8275 cm^2 and 108.9375 cm^2 .

Common Mistakes

X Rounding from the left instead of the right: e.g., rounding 3.14159 to 3 decimal places by looking at the 4th digit.

OK Fix: Only look at the digit immediately after your target precision. Do not round sequentially. For 3.14159 to 3 d.p., look at the 4th digit (5), which is 5 or more, so round up the 3rd digit (1) to 2, giving 3.142.

X Confusing decimal places with significant figures: stating 3.14159 to 3 d.p. is 3.14 (which is actually 2 d.p.).

OK Fix: 3.14159 to 3 d.p. = 3.142 (three digits after the decimal point). The number 3.14 has only 2 decimal places.

Summary

- Round up if the next digit is 5 or more; round down if 4 or less
- Decimal places count digits after the decimal point
- Significant figures count from the first non-zero digit
- Bounds give the interval of possible true values for a rounded measurement
- For calculations with bounds, think about which combination gives the maximum/minimum
- Estimation by rounding to 1 s.f. helps catch calculator errors

A1_Q6: Ratios & Proportion

Hook

A recipe from your grandmother calls for flour and sugar in a 3:2 ratio. You have 500 g of flour — how much sugar do you need? Or you're mixing concrete at a 1:2:4 ratio of cement to sand to gravel — if you need 3.5 m³ of concrete, how much of each ingredient? Ratios are the hidden maths behind every recipe, construction project, and map you've ever used.

Prerequisites

- Basic arithmetic (multiplication and division)
- Simplifying fractions
- Understanding of multiples

Core Explanation

Ratios

A **ratio** compares quantities of the same kind. If there are 3 boys and 2 girls in a group, the ratio is 3:2 (boys to girls).

Simplifying Ratios: Divide all parts by their HCF.

$$12:8 = 3:2 \quad (\text{divide both by } 4)$$

$$20:15:10 = 4:3:2 \quad (\text{divide all by } 5)$$

Sharing in a Given Ratio: To share 60 in the ratio 2:3:

- Total parts: $2 + 3 = 5$
- One part: $60 \div 5 = 12$
- First share: $2 \times 12 = 24$
- Second share: $3 \times 12 = 36$

Increasing in a Ratio: If a recipe for 4 people needs 300 g of flour, to serve 6 people multiply by $\frac{6}{4} = 1.5$. You need $300 \times 1.5 = 450$ g.

Direct Proportion

Two quantities are **directly proportional** if one is a constant multiple of the other:

$$y \propto x \quad (\text{means}) \quad y = kx$$

where k is the **constant of proportionality**.

If y is directly proportional to x , and $y = 12$ when $x = 3$:

$$12 = k \times 3 \Rightarrow k = 4$$

$$y = 4x$$

$$\text{When } x = 7, y = 4 \times 7 = 28.$$

Inverse Proportion

Two quantities are **inversely proportional** if one is a constant divided by the other:

$$y \propto \frac{1}{x} \quad (\text{means}) \quad y = \frac{k}{x}$$

If y is inversely proportional to x , and $y = 6$ when $x = 2$:

$$6 = \frac{k}{2} \Rightarrow k = 12$$

$$y = \frac{12}{x}$$

$$\text{When } x = 4, y = \frac{12}{4} = 3.$$

Unitary Method

Find the value of **one unit** first, then multiply.

$$\text{If 5 apples cost } \$2.50, \text{ then 1 apple costs } \$0.50, \text{ and 8 apples cost } 8 \times \$0.50 = \$4.00.$$

Worked Example

Problem: 3 painters take 8 days to paint a house. How long would 5 painters take if they work at the same rate?

Solution:

This is an **inverse proportion** problem — more painters means fewer days.

Step 1: Find the constant. If p is painters and d is days:

$$p \times d = k$$

$$3 \times 8 = 24$$

$$\text{So } k = 24.$$

Step 2: Use k to find the new value.

$$5 \times d = 24$$

$$d = \frac{24}{5} = 4.8$$

Answer: 5 painters would take 4.8 days (or 4 days and 19.2 hours).

Alternative (unitary method):

One painter takes $3 \times 8 = 24$ days.

Five painters take $24 \div 5 = 4.8$ days.

Common Mistakes

X Adding ratio parts when simplifying: simplifying $10:4$ to $5:4$ (subtracting 6 from the first part)

OK Fix: Always **divide** both parts of a ratio by the same number (their HCF). $10:4 = \frac{10}{2} : \frac{4}{2}$

X Confusing direct and inverse proportion — assuming more means more in every scenario

OK Fix: Ask yourself: if one quantity increases, does the other increase (direct) or decrease (inverse)? More workers means fewer days.

X Forgetting to find the total number of parts when sharing in a ratio

OK Fix: Always add the parts first: for ratio $a:b$, total = $a+b$, one part = total amount $\div (a+b)$.

Summary

- Ratios compare quantities; simplify by dividing all parts by the HCF
- Share in a ratio by finding the value of one part first
- Direct proportion: $y = kx$ — as one goes up, the other goes up
- Inverse proportion: $y = \frac{k}{x}$ — as one goes up, the other goes down
- The unitary method: find one unit, then multiply
- Use the constant of proportionality k to solve proportion problems

A1_Q7: Indices & Standard Form

Hook

How many atoms are there in a grain of sand? Roughly 3×10^{18} — that's 3000000000000000000, a number so large it's meaningless to write out all the zeros. And the mass of a proton is 1.67×10^{-27} kg — a decimal so tiny it would take a full line just to write its zeros. Indices and standard form are how scientists and engineers tame the universe's extremes.

Prerequisites

- Basic multiplication and division
- Understanding of powers (e.g., $2^3 = 8$)
- Fraction arithmetic
- Decimal place value

Core Explanation

Laws of Indices

An **index** (plural: indices) or **power** shows how many times a number is multiplied by itself.

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$a^0 = 1$$

$$a^{-n} = \frac{1}{a^n}$$

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

Examples:

$$- 2^3 \times 2^4 = 2^{3+4} = 2^7 = 128$$

$$- 5^6 \div 5^2 = 5^{6-2} = 5^4 = 625$$

$$- (3^2)^3 = 3^{2 \times 3} = 3^6 = 729$$

$$- 7^0 = 1$$

$$- 3^{-2} = \frac{1}{3^2} = \frac{1}{9}$$

$$- 8^{\frac{2}{3}} = (\sqrt[3]{8})^2 = 2^2 = 4$$

$$- 16^{-\frac{3}{4}} = \frac{1}{16^{\frac{3}{4}}} = \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \frac{1}{8}$$

Standard Form (Scientific Notation)

A number in standard form is written as:

$$A \times 10^n$$

where $1 \leq A < 10$ and n is an integer.

Large numbers:

$$- 4500000 = 4.5 \times 10^6 \text{ (move decimal 6 places left)}$$

- $3.2 \times 10^8 = 320000000$ (move decimal 8 places right)

Small numbers:

- $0.0000072 = 7.2 \times 10^{-6}$ (move decimal 6 places right, negative exponent)

- $4.1 \times 10^{-3} = 0.0041$ (move decimal 3 places left)

Calculations in standard form:

$$(3 \times 10^5) \times (2 \times 10^3) = 6 \times 10^8$$

$$(8 \times 10^7) \div (2 \times 10^3) = 4 \times 10^4$$

$$(5 \times 10^4) + (3 \times 10^4) = 8 \times 10^4$$

$$(3 \times 10^5) + (4 \times 10^4) = 3 \times 10^5 + 0.4 \times 10^5 = 3.4 \times 10^5$$

Worked Example

Problem: Simplify $\frac{27^{\frac{2}{3}} \times 9^{-1}}{81^{\frac{1}{2}}}$ and express your answer in the form 3^n .

Solution:

Step 1: Write all terms as powers of 3.

$$27 = 3^3, \quad 9 = 3^2, \quad 81 = 3^4$$

Step 2: Substitute.

$$\frac{(3^3)^{\frac{2}{3}} \times (3^2)^{-1}}{(3^4)^{\frac{1}{2}}}$$

Step 3: Use $(a^m)^n = a^{mn}$.

$$\frac{3^{3 \times \frac{2}{3}} \times 3^{2 \times (-1)}}{3^{4 \times \frac{1}{2}}} = \frac{3^2 \times 3^{-2}}{3^2}$$

Step 4: Add exponents in numerator.

$$\frac{3^{2 + (-2)}}{3^2} = \frac{3^0}{3^2}$$

Step 5: Use $a^0 = 1$ and $a^m \div a^n = a^{m-n}$.

$$\frac{1}{3^2} = 3^{-2}$$

Answer: 3^{-2}

Common Mistakes

X Adding exponents when multiplying different bases: $2^3 \times 3^2 = 6^5$

OK Fix: The index law $a^m \times a^n = a^{m+n}$ only works when the **bases are the same**. $2^3 \times 3^2 = 8 \times 9 = 72$

X Treating negative exponents as making the answer negative: $2^{-3} = -8$

OK Fix: Negative exponents mean **reciprocals**, not negative numbers. $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$

X Forgetting to adjust the coefficient when adding/subtracting numbers in standard form with different powers: $3 \times 10^5 + 4 \times 10^3 = 300 \times 10^3 + 4 \times 10^3 = 304 \times 10^3$

OK Fix: You must convert to the same power of 10 first. $3 \times 10^5 + 4 \times 10^3 = 300 \times 10^3 + 4 \times 10^3 = 304 \times 10^3 = 3.04 \times 10^5$

Summary

- When multiplying, add exponents (same base): $a^m \times a^n = a^{m+n}$
- When dividing, subtract exponents: $a^m \div a^n = a^{m-n}$
- Power of a power: $(a^m)^n = a^{mn}$
- Anything to power 0 equals 1: $a^0 = 1$
- Negative exponents mean reciprocal: $a^{-n} = \frac{1}{a^n}$
- Fractional exponents mean roots: $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
- Standard form: $A \times 10^n$ where $1 \leq A < 10$
- Adjust coefficients to the same power when adding/subtracting in standard form